

Chemical & Water Desalination Engineering Program

Study Plan Booklet

Academic Year 2021-2022

Semester by Semester Study Plan	_____	Page 1
Flow Chart Study Plan	_____	Page 2
Technical Electives List	_____	Page 3
University Electives List	_____	Page 4

Chemical & Water Desalination Engineering Program
Academic Year 2021-2022

Semester-by-semester Study Plan
(131 Total Credit Hours - 52 Total Courses)

University Requirement	(24 Credit Hours)
College Requirement	(26 Credit Hours)
Program Requirement	(81 Credit Hours)

Course Number Breakdown				
Mass Transfer	College Code	Department Code	Level	Course Code
0408 278	04	08	2	78

First Year (Freshman Year)				
First Semester			Second Semester	
0104 100	Islamic Culture	3	0202 102	Arabic Language 3
0202 112	English for Academic Purposes	3	1501 100	Introduction to I.T. 3
1430 115	Physics 1	3	1440 161	Calculus II for Engineering 3
1430 116	Physics 1 Lab	1	0202 207	Technical Writing 3
1420 101	General Chemistry (1)	3	1430 117	Physics 2 3
1440 133	Calculus I for Engineering	3	0408 170	Intro. to Chem and Env Sci. & Eng. 2
1420 102	General Chemistry I Lab	1		
	Total Credit Hours	17		Total Credit Hours 17
Second Year (Sophomore Year)				
Third Semester			Fourth Semester	
0408 270	Water Resources and Management	3	0408 276	Material Processing Engineering 3
0408 271	Water Chemistry and Analysis	3	0408 277	Principles of Heat Transfer 3
0408 272	Chemical Thermodynamics	3	0408 278	Mass Transfer 3
0408 273	Applied Fluid Mechanics	3	0408 279	Polymer Engineering 3
0408 274	Membrane I	2	0408 280	Principles of Water Treatment Tech. 3
0408 275	Fund. of Thermal Desalination Proc.	2	0408 281	Chemical Thermodynamics (Lab) 1
			0408 282	Applied Fluid Mechanics (Lab) 1
	Total Credit Hours	16		Total Credit Hours 17
Third Year (Junior Year)				
Fifth Semester			Sixth Semester	
0204 102	UAE Society	3	xxxx xxx	University Elective I 3
1440 261	Differential Equations for Eng.	3	0401 301	Engineering Economics 3
0408 370	Thermal Sys. Design and Analysis	3	0405 221	Eng. Probability & Statistics 3
0408 371	Membrane II	3	0408 375	Membrane III 3
0408 372	Pre & Post Treat. for Desal. Plants	3	1420 103	General Chemistry 2 3
0408 373	Polymer Eng lab	1	0408 377	Principles of Water Treatment Tech. (Lab) 1
1502 300	Prof. & Social Issues in Engineering	1	0408 378	Heat and mass Transfer (Lab) 1
	Total Credit Hours	17		Total Credit Hours 17
Summer Semester				
0408497	Practical Training (240 working hours in 6 - 8 weeks)	0		
Fourth Year (Senior Year)				
Seventh Semester			Eighth Semester	
0408 470	Process Control and Simulation	3	0408 479	Senior Design Project II 3
0408 471	Plant Design and Cost Analysis	3	0408 xxx	CWDE Technical Elective II 3
0420 104	General Chemistry 2 Lab	1	0408 xxx	CWDE Technical Elective III 3
0408 383	Desalination Lab	1	0302 200	Fund. of Innovations & Entrepreneur 3
0408 474	Water and Energy	3	xxxx xxx	University Elective II 3
0408 478	Senior Design Project I	1		
0408 xxx	CWDE Technical Elective I	3		
	Total Credit Hours	15		Total Credit Hours 15
TOTAL CREDIT HOURS = 131				

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Study Plan Flowchart

University Requirement (24 Credit hours)

College Requirement (26 Credit hours)

Program Requirement (81 Credit hours)

Course A → Course B

Course A → Course B

131 Total Credit Hours
52 Total Courses

Course A is a Pre-requisite to course B

Course A is a Pre/Co-requisite to course B

Subject

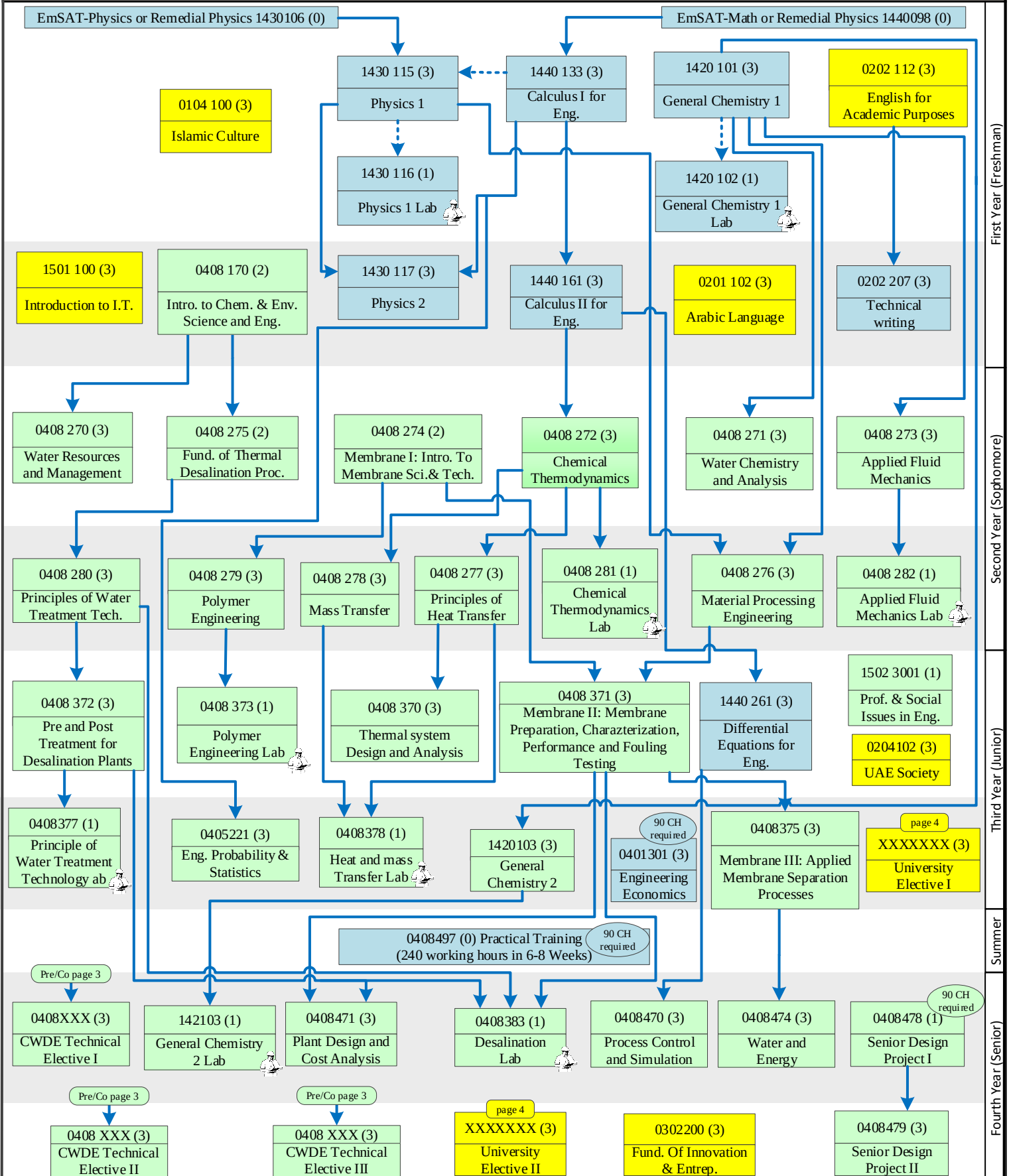
0408 423 (1)

Course Number

Credit hours

Course name

Water Desalination Lab



Chemical & Water Desalination Engineering Program
Academic Year 2021-2022
List of Technical Electives

#	Course Code	Course Name	Pre-requisite	Credit Hours
1	0408 374	Thermal Desalination Process Design and Analysis	None	3
2	0408 376	Sea Water Desalination using Non-Conventional Methods	None	
3	0408 480	Renewable Energy Desalination	None	
4	0408 481	Hybrid Desalination Processes	None	
5	0408 482	Modeling & Simulation of Multi-Stage Flash Desalination	None	
6	0408 483	Membrane Distillation	None	
7	0408 484	Pressure Retarded Osmoses (PRO) Membranes	None	
8	0408 485	Nanotechnology for Water Treatments and Desalination	None	
9	0408 486	Nuclear Desalination	None	
10	0408 487	Special Topic	None	
11	0408 489	Emerging Membrane Processes	None	
12	0408 493	Introduction to Biochemical Engineering	None	
13	0408 494	Oil and Gas Technology	None	

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List of University Electives

Important Notes:

You must select one course only from each of the below 2 domains of courses. If you select two courses from the same group, only one course will be considered for you.

Domain 1 (Humanities Social and Arts)			
#	Course Code	Course Name	Credit Hours
1	0103 103	Islamic System	3
2	0103 104	Prof. Ethics in Islamic Sharia	
3	0104 130	Analytical Biog of the Prophet	
4	0201 140	Intro. to Arabic Literature	
5	0202 130	French Language	
6	0203 100	Islamic Civilization	
7	0203 102	History of the Arabian Gulf	
8	0203 200	Hist of Sciences among Muslims	
9	0204 103	Principles of Sign Language	
10	0206 102	Fundamentals/Islamic Education	
11	0206 103	Introduction to Psychology	
12	0302 150	Intro.to Bus for Non-Bus	
13	0308 131	Personal Finance	
14	0308 150	Intro to Economics (Non-B)	
15	0602 246	Human Rights in Islam	
16	0700 310	Intro to Islamic Art & Design	
17	0710 109	Arts and Medicine	
18	0800 107	Media in Modern Societies	
19	0900 107	History of Medical and H.Sc.	

Domain 2 (Natural & Health Sciences)			
#	Course Code	Course Name	Credit Hours
1	0401 142	Man and The Environment	3
2	0406 102	Introduction to Sustainability	
3	0503 101	Health and Safety	
4	0505 101	Fitness and Wellness	
5	0507 101	Health Awareness and Nutrition	
6	1430 101	Astro & Space Sciences	
7	1450 100	Biology and Society	
8	1502 233	Introduction to AI	